"The Cluck is Ticking!"

(Our iGEM 2026 project website as summarized by AI in layman terms)

"The Cluck is Ticking!", a living bacterial sensor and smartphone-friendly model that measures real-time meat spoilage so people do not throw away safe chicken or eat bad meat. Instead of relying on static "sell-by" dates, their system detects a specific chemical released as bacteria rot poultry and translates that signal into a color change and an exact remaining shelf life.

The Core Problem

• Static dates fail. Printed expiration dates are rough guesses that ignore how warm or cold meat actually stayed during shipping and storage.

• Massive waste and risk. Guessing leads consumers and stores to throw out billions of dollars of safe poultry while still missing early bacterial growth.

How the Biology Works

• The Target Chemical (Cadaverine). When bacteria break down amino acids in meat and poultry, they produce a foul-smelling compound called cadaverine. High cadaverine means the meat is spoiling.

• The Sensor Protein (CadC). The team engineered harmless E. coli bacteria to use a natural membrane sensor called CadC. CadC acts like a chemical tripwire that senses cadaverine outside the cell and sends a signal inside to turn genes on or off.

• The Visual Reporter (Chromoproteins). Instead of using fluorescent proteins that require expensive UV lights to see, they wired the sensor to amilCP, a coral-derived chromoprotein that turns the bacteria bright blue in normal room light.

The Engineering Pivot

• The Unexpected Twist. In their first lab tests (testing 0, 4, and 40 mM of cadaverine), their initial genetic circuit worked backward. Instead of turning blue when cadaverine appeared, the bacteria started blue and faded to clear as cadaverine increased because high cadaverine inhibited their initial CadC setup.

• The Redesign. Rather than hiding the quirk, they ran a second engineering cycle and designed four next-generation plasmids. They added a genetic "NOT gate" using a cI repressor (so high cadaverine turns off the repressor and turns on the color) alongside a red protein tag (mRFP) to tune the color intensity.

• Color Tuning. They also used site-directed mutagenesis to tweak single amino acids in their chromoprotein, creating a palette of different shades to make subtle spoilage stages easier to tell apart.

The Math, Hardware, and App Pipeline

• Computer Vision Calibration. Because human eyes judge color differently depending on lighting, the team built a software pipeline that reads camera pixel data and matches it against lab spectrophotometer (plate reader) measurements.

• The Hill Model. They plugged that color intensity into a Hill activation equation, which converts the shade of blue into an exact cadaverine concentration.

• Shelf-Life Countdown. Finally, a bacterial growth and spoilage kinetics model uses that current cadaverine level to calculate how many hours or days of safe shelf life remain.

iGEM Video Link: https://video.igem.org/w/1qUXaL9ND5KZCcmvqKZeNL